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Algorithm DC_CLEVER_MULTIPLICATION(A, B)
// Description : Perform multiplication of large numbers using divide and conquer strategy.
// Input : Number A and B, where  $A = a_{n-1}...a_1a_0$ , and  $B = b_{n-1}...b_1b_0$ 
// Output : Multiplication of A and B as C, i.e.  $C = A * B$ 

if  $|a| == 1$  or  $|b| == 1$  then
    return  $A * B$ 
else
     $n \leftarrow \max(|A|, |B|)$ 
     $c_2 \leftarrow \text{DC\_CLEVER\_MULTIPLICATION}(a_1, b_1)$ 
     $c_0 \leftarrow \text{DC\_CLEVER\_MULTIPLICATION}(a_0, b_0)$ 
     $x \leftarrow \text{DC\_CLEVER\_MULTIPLICATION}(a_1 + a_0, b_1 + b_0)$ 
    return  $c_2 * 10^n + (x - c_2 - c_0) * 10^{n/2} + c_0$ 
end

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